

AI-Powered Fish Monitoring — Full Roadmap

AI-Powered Fish Monitoring System — Full Roadmap (Including Week 1)

This roadmap summarizes the entire plan from MVP (motion-based) to AI-powered detection and productization.

■ Goal

Build a monitoring system that:

- Tracks **7 individual goldfish**
- Logs movement/activity metrics to **InfluxDB**
- Visualizes & alerts in **Grafana**
- Starts with **motion detection (MVP)**, then upgrades to **AI detection/tracking**

Phase 1 — Motion-Based MVP (Weeks 1-4)

Week 1 — Environment & Camera Setup (■ Completed)

- **Flash Raspberry Pi OS** and set up the Pi (network, SSH)
- **Install core tools:** Python 3, OpenCV, Git
- **Install & start** InfluxDB + Grafana
- **Connect & verify camera:** Camera Module 3 (IMX708) using `rplicam-hello`
- **Stabilize lighting & mounting** over the tank
- **Outcome:** Pi + camera + observability stack ready

Week 2 — Motion Detection & Basic Tracking

- Implement OpenCV-based **background subtraction** (MOG2)
- Detect motion -> **contours** -> **centroids**
- Add a **simple multi-target tracker** to keep temporary IDs
- Compute **per-minute activity** per fish (distance moved, activity index)
- Push metrics to **InfluxDB**

Week 3 — Dashboard & Alerts

- Create Grafana panels (one per fish + an overview)
- Configure **alerts** (e.g., inactivity > 10 min vs baseline)
- Tune thresholds & filters (ignore bubbles/reflections)

Week 4 — Calibration & Baselines

- Run **5-7 days** continuously and collect baselines per fish
- Adjust sensitivity/thresholds & lighting angles
- Document "normal" vs "anomalous" behavior

Phase 2 — AI Upgrade (Weeks 5-10)

Weeks 5-6 — Model & Hardware

- Add **Jetson Nano** or **Coral USB TPU** (if staying on Pi)
- Validate **YOLOv8n / EfficientDet** inference performance
- Decide on runtime (OpenCV DNN / TensorRT / TFLite)

Weeks 7-8 — AI Detection + Tracking

- Replace contours with **AI detections (boxes)** per frame
- Integrate a tracker (e.g., **Norfair / Deep SORT** or nearest-neighbor)
- Compare accuracy with Week 2 motion method

Weeks 9-10 — Health Insights

- Add **zone** (top/bottom) + **orientation** (sideways float) signals
- Enrich alerts with context (posture, lingering, sudden stops)
- Start labeling events for future fine-tuning

Phase 3 — Productization (Weeks 11-16)

- Package as **Docker** or ready-made SD image
- Add a **setup wizard** and a lightweight web UI
- Remote update channel for **model/threshold** changes
- Prepare documentation & basic support scripts

Running Glossary (Metrics)

- **distance_per_min**: pixel or cm distance moved per minute

- **activity_index**: % frames with motion per minute
- **zone_time_top/bottom**: time spent near surface or substrate
- **alert_low_activity**: rule that triggers if a fish deviates from baseline

Notes

- Keep **frame processing** modest (e.g., 1280×720 @ 15-30 fps)
- Prefer **fixed camera** and **consistent lighting** to reduce false motion
- Log **once per minute** to reduce DB load